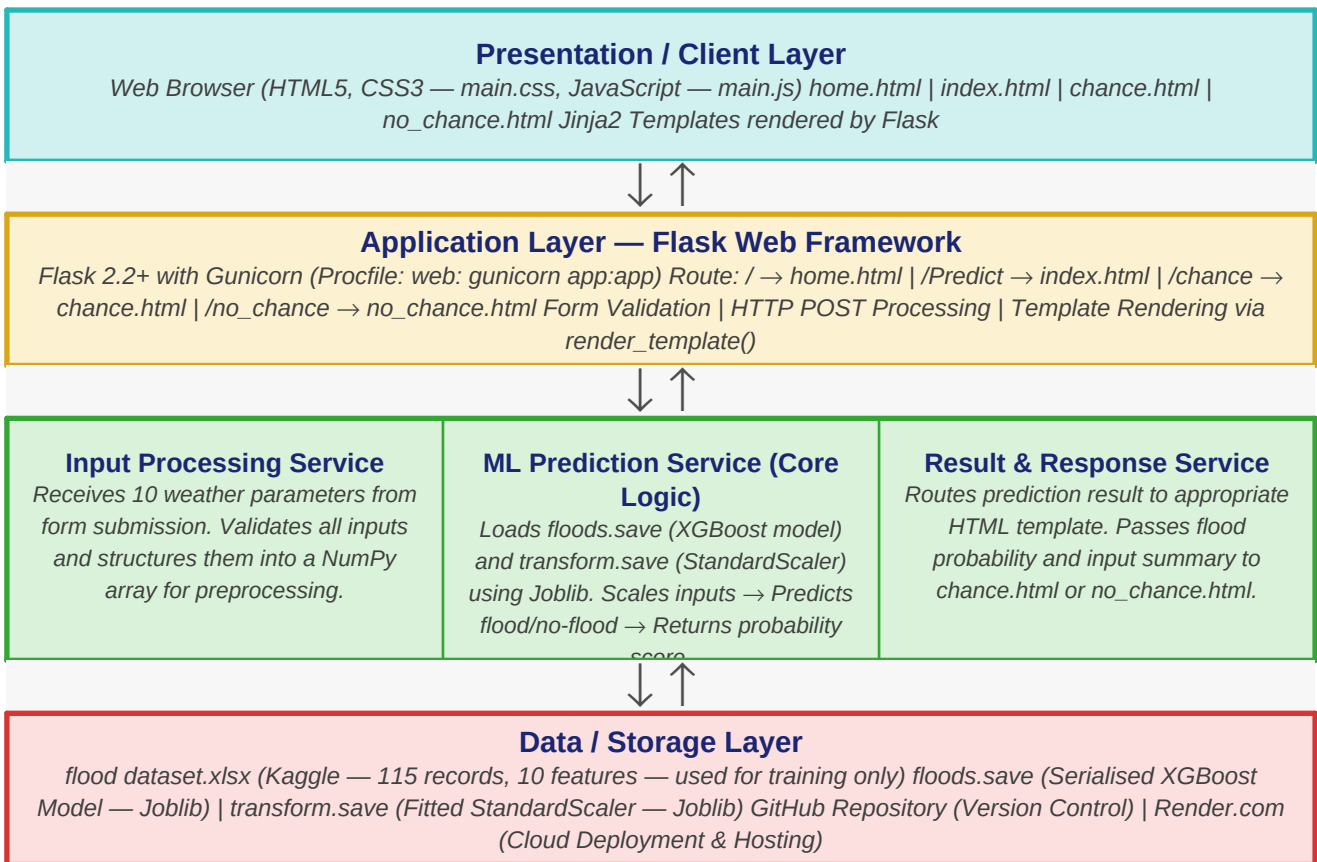


Solution Architecture

Date	26/06/2026
Team ID	Not Assigned
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of the Rising Waters solution architecture. The diagram maps out the Presentation Layer, Application Layer, and Data Layer with specific technologies and components used in this flood prediction system.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Delivers the user interface to the browser. Renders HTML pages using Jinja2 templates. Provides input form for weather parameters, displays flood/no-flood results with probability scores and advisory actions.	HTML5, CSS3 (main.css), JavaScript (main.js), Jinja2 Templates
Application Layer (API Gateway)	Flask web framework handles all HTTP routing, form data processing, and template rendering. Gunicorn serves as the production WSGI server. Routes: /, /Predict, /chance, /no_chance.	Python 3.8+, Flask 2.2+, Gunicorn, Procfile

Component Name	Description / Role in Architecture	Technologies Used
Input Processing Service	Receives 10 meteorological parameters via HTTP POST form submission. Validates all inputs, structures them into a NumPy array, and passes to the ML prediction service for scaling and inference.	Python, NumPy, Flask request.form
ML Prediction Service (Core Logic)	Loads the trained XGBoost model (floods.save) and fitted StandardScaler (transform.save) using Joblib. Scales the input array and generates a binary flood prediction with probability score.	XGBoost, Scikit-learn, Joblib, NumPy (floods.save, transform.save)
Result & Response Service	Routes the prediction output to the appropriate result page. Passes flood probability percentage and input summary dictionary to chance.html (flood) or no_chance.html (no flood).	Flask render_template(), Jinja2, chance.html, no_chance.html
Data / Storage Layer	Stores the training dataset (flood dataset.xlsx), serialised ML model (floods.save), and fitted scaler (transform.save). GitHub provides version control and Render.com provides cloud hosting.	flood dataset.xlsx (Kaggle), Joblib .save files, GitHub, Render.com